

Optimal Pin-Point Positioning via SLAM Correspondence: Continued Fractions and Elliptic Integrals on Non-Euclidean Manifolds

Cartik Sharma^{1,2*}, NeuroMorph Quantum Systems Division¹

¹Institute for Computational Neuroimaging & Robotic Surgery

²Centre for Advanced Surgical Robotics

Abstract

Simultaneous Localization and Mapping (SLAM) for surgical robotics demands exceptional sub-millimeter positioning accuracy. Modern approaches mapping sensor coordinates across varying magnetic flux and environmental perturbations operate on non-Euclidean manifolds. We demonstrate the injection of Jacobi elliptic integrals of the first kind alongside Stern-Brocot continued fraction expansions to govern SLAM correspondence, overcoming Euclidean error drifts.

1. Introduction to Manifold Kinematics

Traditional Extended Kalman Filters and position mapping rely on linear estimations. Surgical environments present high-curvature metric distortions. The correspondence between SLAM anchor points and sensor readings dictates tracking accuracy.

2. Elliptic Integral Mapping

The non-Euclidean deformation of the spatial metric is governed by the Jacobi incomplete elliptic integral of the first kind $F(\phi, k)$:

$$F(\phi, k) = \int_0^\phi \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}} \quad (1)$$

Elliptic amplitude function binding spatial distortions

$$\phi = \arcsin(z / ||p||), k = ||p|| / 2 \quad (2)$$

Tracking amplitude and modulus

3. Continued Fraction Approximation

Discrete jump resonances occur when aligning raw localized bounds to precise sub-millimeter targets. Using continued fractions provides rational approximations bounding the irrational spatial gap δ .

$$z_{\text{residual}} = a_n + 1 / (a_{n+1} + 1 / (a_{n+2} + \dots)) \quad (3)$$

SLAM residual continued fraction

$$\delta_{\text{correction}} = |z_{\text{residual}} - p_n / q_n| \quad (4)$$

Fractal residual offset correction

4. Optimal Pin-Point Positioning Law

Combining the non-Euclidean bound tracking from elliptic integrals and discrete SLAM correspondence anchors bounded by fractal correction gives our target tracking law:

$$P_{\text{err}} = ||p_{\text{true}} - p_{\text{target}}|| + 0.01 \times \delta_{\text{correction}} \quad (5)$$

Adjusted positioning loss function